



King Abdul Aziz University

Faculty of Computing & Info Tech. Computer Science Department

First Exam (19-08-1443 / 22-03-2022)

CPCS-202 (Programming 1)

Term: (Spring 2022), Time: 80 Minutes

Student ID: _____ Section: _____

Student Name: _____

Question	Max. Mark	Obtained Mark	Total Mark
I	20		
II	40		
III	40		

Instructor:

- ☐ Dr. Abdulaziz Alnori
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Part I: Multiple Choice (20 points)

Circle the correct answer in the following multiple-choice questions:

1. Which of the following is a correct example of typecasting? (3 marks)
 - a. `int a = (int) 56.75;`
 - b. `int a = (float) 56.75;`
 - c. `float a = 56.75;`
 - d. `double a = 56;`

2. Which of the following is a not a numerical data type? (3 marks)
 - a. `short`
 - b. `char`
 - c. `long`
 - d. `byte`

3. Which of the following is a wrong declaration of a variable? (3 marks)
 - a. `double payment;`
 - b. `short $age11;`
 - c. `int 5cm;`
 - d. `long cm2mile;`

4. What will be displayed by the following switch statement: (3 marks)

```
int b = 1;
switch(b) {
    case 1: System.out.print(b);
    case 3: System.out.print(b);
    case 5: System.out.print('b'); break;
    case 6: System.out.println("Five");
    break;
}
```

 - a. 11b
 - b. bbbFive
 - c. 1
 - d. b

5. If x=10 and y=2, which of the following expression evaluates to *true*? (4 marks)
 - a. `(x > 0) && (y > 3)`
 - b. `(x < 0) || (y >=1)`
 - c. `(x < 0) || (y == 1)`
 - d. `!(x < 0) && (y > 2)`

6. The following expression `int x = 3+4*3+2;` results in x having the value: (4 marks)
 - a. 35
 - b. 23
 - c. 17
 - d. None of the above

Part II (40 points)

Q1. (20 points) Translate the following algorithm into Java code:

CLO-2

1. Ask a user to enter five phone numbers (which consist of exactly 12 digits) from the user.
2. For each number entered, the program should print the name of the country based on the first three digits of the phone number entered as follows
 - a. 966 - Saudi Arabia
 - b. 971 - United Arab Emirates
 - c. 973 - Bahrain
 - d. 968 - Oman
3. If the first three digits do not match any of the four codes mentioned above, the program should print "Code unknown"

(Note : The phone number should be read as a number and not as a String. Also the number should not be converted into a String and String or charAt() methods should not be used to solve this problem)

Sample input and output

```
Please enter a 12 digit phone number : 966325874103 ↵
Saudi Arabia
Please enter a 12 digit phone number : 966582471968 ↵
Saudi Arabia
Please enter a 12 digit phone number : 968352987412 ↵
Oman
Please enter a 12 digit phone number : 745869321561 ↵
Code Unknown
Please enter a 12 digit phone number : 971369852741 ↵
United Arab Emirates
```

Solution

```
import java.util.Scanner;

public class MyClass {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
```

```
}  
}
```

Q2. 20 points)

CLO-12

Write the outputs of the following code segments in the boxes provided

```
1)  int B = 5, C = 4;  
    System.out.println( B + B + "B");  
    System.out.println(C + ('B' + B) + B);  
    System.out.println(B + "B" + B);  
    System.out.println(C + "C" + 'B');
```

(Hint: the ASCII value of 'A' is 65)

```
2) char num = '3';  
   int num2 = 5;  
   switch(num2){  
       case 1 :System.out.print(num + num2);  
           break;  
       case 5 :System.out.print(num2);  
       case 2 :System.out.print(num);  
       case 4 :System.out.println("num");  
       default:System.out.println("Complete");  
   }
```

```
3) int i, j = 6, k = 2, n = 1;  
   for (i = j - k; i < j * 2; i++){  
       if(i > j + 3 || i % 3 == 0)  
           continue;  
       n = i + i;  
       if (i == 10)  
           break;  
   }  
   System.out.println("the result is " + n);
```

Part III: (40 points)

Q1. (10 points)

Write a program that randomly generates a number ($2 \leq x \leq 8$) between 2 and 8, inclusive. The program should repeatedly ask the user to enter a number (y) until the user enters a number larger than the random number ($y \geq x$). For each number entered (y) that is smaller than x, the program should display the message "*Enter a number larger than $\underline{y}$ but smaller than or equal to 8*". At the end, print a message "*Great!! Your number $\underline{y}$ is larger than or equal to $\underline{x}$* "

Sample Run

```
Enter a number larger than 2 but smaller than or equal to 8 : 3 ↵
Enter a number larger than 3 but smaller than or equal to 8 : 5 ↵
Enter a number larger than 5 but smaller than or equal to 8 : 6 ↵
Enter a number larger than 6 but smaller than or equal to 8 : 7 ↵
```

Great!! Your number 7 is larger than or equal to 7

Solution

```
import java.util.Scanner;

public class MyFirstProgram {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in) ;
```

```
    }
}
```

Q2. (30 points)

Write a program that reads a string of length 10 from the user. It should then search if the letters 'A', 'c' and 'e' (in the same order) appear anywhere in the string. Your program should then display "*Found*" if the three characters are found in the same order. Otherwise the program should display "*Not found*".

Note : assume the string entered is always of length 10.

CLO-7, 8

Sample run #1:

Enter a string : **HelloWorld** ↵
Not found!

Sample run #2:

Enter a string : **HAecloWorl** ↵
Not found!

Sample run #3:

Enter a string : **HelWorAced** ↵
Found!

Sample run #4:

Enter a string : **ellAWorcle** ↵
Found!

Solution

```
import java.util.Scanner;

public class MyFirstProgram {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in) ;
```

```
    }
}
```